

NAME OF THE MEDICINAL PRODUCT

MOXICUM 400 mg/250 ml solution for infusion

QUALITATIVE AND QUANTITATIVE COMPOSITION

1 vial of 250ml contains 436.25mg Moxifloxacin hydrochloride equivalent to 400mg Moxifloxacin

PHARMACEUTICAL FORM

Solution for infusion

Clear, greenish yellow solution practically free from particles

CLINICAL PARTICULARS**Therapeutic indications**

MOXICUM 400 mg/250 ml solution for infusion is indicated for the treatment of adults (≥ 18 years of age) with the following bacterial infections caused by susceptible strains:

- Community acquired pneumonia
- Complicated skin and skin structure infections
- Complicated intraabdominal infections including polymicrobial infections such as abscesses

Consideration should be given to official guidance on the appropriate use of antibacterial agents.

MOXICUM should be only used:

- When *Pseudomonas fluorescens* is considered AND the patient is allergic to antipseudomonal penicillins/ cephalosporin; or
- For resistant organism with no other alternative antibiotics available (see section “*Pharmacodynamic properties*” for susceptibility data).

MOXICUM should not be used >24 hours post operation.

Posology and method of administration**Dose (adults)**

The recommended dose for MOXICUM is 400 mg once daily (1 film-coated tablet or 250 mL solution for infusion, respectively) for the above-mentioned indications and should not be exceeded.

Duration of treatment

The duration of treatment should be determined by the severity of the indication or clinical response. The following general recommendations for the treatment of upper and lower respiratory tract infections are made:

Solution for infusion:

Therapy may be initial intravenous administration, followed by oral administration of film-coated tablets when clinically indicated.

Pneumonia: community acquired pneumonia: The recommended total treatment duration for sequential administration (intravenous followed by oral therapy) is: 7-14 days

Complicated skin and skin structure infections total treatment duration for sequential therapy (intravenous followed by oral therapy): 7 – 21 days

Complicated intraabdominal infections total treatment duration for sequential therapy (intravenous followed by oral therapy): 5 – 14 days

The recommended duration of treatment for the indication being treated should not be exceeded.

MOXICUM have been studied in clinical trials for up to 21 days (in complicated skin and skin structure infections).

Method of administration

Solution for infusion:

The solution for infusion should be infused intravenously over 60 minutes.

A statement “Infuse over a period of 60 minutes” should be given on the primary packaging material of the infusion solution.

The solution for infusion can be administered directly or via a T-tube together with compatible infusion solutions.

The following co-infusions were found to form stable mixtures over a period of 24 hours at room temperature with MOXICUM, and can therefore be considered as compatible with MOXICUM:

- Water for Injections
- Sodium Chloride 0.9%
- Sodium Chloride 1 molar
- Glucose 5%
- Glucose 10%
- Glucose 40%
- Xylitol 20%
- Ringer’s Solution
- Lactated Ringer’s Solution

If MOXICUM for infusion is to be given with another drug, each drug should be given separately. Only clear solutions are to be used.

Missed dose

If a dose is missed, it should be taken anytime but not later than 8 hours prior to the next scheduled dose.

If less than 8 hours remain before the next dose, the missed dose should not be taken and treatment should be continued as prescribed with the next scheduled dose. Double doses should not be taken to compensate for a missed dose.

Geriatric patients

No adjustment of dosage is required in the elderly.

Pediatric patients

The efficacy of MOXICUM in children and adolescents has not been established. No recommendation on posology can be made. The safety of MOXICUM in children below the age of 6 years has not been established.

Ethnic differences

No adjustment of dosage is required in ethnic groups.

Patients with hepatic impairment

No dosage adjustment is required in patients with impaired liver function

Patients with renal impairment

No dose adjustment is required in patients with renal impairment (including creatinine clearance ≤ 30 mL/min/1.73m²) and in patients on chronic dialysis i.e. hemodialysis and continuous ambulatory peritoneal dialysis.

Contraindications

Moxifloxacin is contraindicated in below mentioned conditions:

- Known hypersensitivity to any component of Moxifloxacin or other quinolones or any of the excipients.
- Pregnancy and lactation.
- Patients below 18 years of age.

Special warnings and precautions for use

The use of moxifloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section *Undesirable Effects*). Treatment of these patients with moxifloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Moxifloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Prolongation of QTc interval and potentially QTc-prolongation-related clinical conditions

Moxifloxacin has been shown to prolong the QTc interval on the electrocardiogram in some patients. The magnitude of QT prolongation may increase with increasing plasma concentrations due to rapid intravenous infusion. Therefore, the duration of infusion should not be less than the recommended 60 minutes and the intravenous dose of 400 mg once a day should not be exceeded.

Treatment with moxifloxacin should be stopped if signs or symptoms that may be associated with cardiac arrhythmia occur during treatment, with or without ECG findings.

Moxifloxacin should be used with caution in patients with any condition pre-disposing to cardiac arrhythmias (e.g. acute myocardial ischaemia) because they may have an increased risk of developing ventricular arrhythmias (incl. torsade de pointes) and cardiac arrest.

Moxifloxacin should be used with caution in patients who are taking medications that can reduce potassium levels. Moxifloxacin should be used with caution in patients who are taking medications associated with clinically significant bradycardia.

Female patients and elderly patients may be more sensitive to the effects of QTc-prolonging medications such as moxifloxacin and therefore special caution is required.

Hypersensitivity/allergic reactions

Hypersensitivity and allergic reactions have been reported for fluoroquinolones including moxifloxacin after first administration. Anaphylactic reactions can progress to a life-threatening shock, even after the first administration. In cases of clinical manifestations of severe hypersensitivity reactions moxifloxacin should be discontinued and suitable treatment (e.g. treatment for shock) initiated.

Severe liver disorders

Cases of fulminant hepatitis potentially leading to liver failure (including fatal cases) have been reported with moxifloxacin. Patients should be advised to contact their doctor prior to continuing treatment if signs and symptoms of fulminant hepatic disease develop such as rapidly developing asthenia associated with jaundice, dark urine, bleeding tendency or hepatic encephalopathy.

Liver function tests/investigations should be performed in cases where indications of liver dysfunction occur.

Serious bullous skin reactions

Cases of bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis have been reported with moxifloxacin. Patients should be advised to contact their doctor immediately prior to continuing treatment if skin and/or mucosal reactions occur.

Patients predisposed to seizures

Quinolones are known to trigger seizures. Use should be with caution in patients with CNS disorders or in the presence of other risk factors which may predispose to seizures or lower the seizure threshold. In case of seizures, treatment with moxifloxacin should be discontinued and appropriate measures instituted.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesias, hypoaesthesias, dysaesthesias, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with moxifloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of an irreversible condition (see section *Undesirable Effects*).

Psychiatric reactions

Psychiatric reactions may occur even after the first administration of quinolones, including moxifloxacin. In very rare cases depression or psychotic reactions have progressed to suicidal thoughts and self-injurious behaviour such as suicide attempts. In the event that the patient develops these reactions, moxifloxacin should be discontinued and appropriate measures instituted. Caution is recommended if moxifloxacin is to be used in psychotic patients or in patients with history of psychiatric disease.

Antibiotic-associated diarrhoea incl. colitis

Antibiotic-associated diarrhoea (AAD) and antibiotic-associated colitis (AAC), including pseudomembranous colitis and Clostridium difficile-associated diarrhoea, has been reported in association with the use of broad spectrum antibiotics including moxifloxacin and may range in severity from mild diarrhoea to fatal colitis. Therefore it is important to consider this diagnosis in patients who develop serious diarrhoea during or after the use of moxifloxacin. If AAD or AAC is suspected or confirmed, ongoing treatment with antibacterial agents, including moxifloxacin, should be discontinued and adequate therapeutic measures should be initiated immediately. Furthermore, appropriate infection control measures should be undertaken to reduce the risk of transmission. Drugs inhibiting peristalsis are contraindicated in patients who develop serious diarrhoea.

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur up to several months after discontinuation of therapy. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with moxifloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilization). Corticosteroids should not be used if signs of tendinopathy occur. .

Patients with renal impairment

Elderly patients with renal disorders should use moxifloxacin with caution if they are unable to maintain adequate fluid intake, because dehydration may increase the risk of renal failure.

Vision disorders

If vision becomes impaired or any effects on the eyes are experienced, an eye specialist should be consulted immediately

Dysglycemia

As with all fluoroquinolones, disturbances in blood glucose, including both hypoglycemia and hyperglycemia have been reported with moxifloxacin. In moxifloxacin-treated patients, dysglycemia occurred predominantly in elderly diabetic patients receiving concomitant treatment with an oral hypoglycemic agent (e.g. sulfonylurea) or with insulin. In diabetic patients, careful monitoring of blood glucose is recommended.

Prevention of photosensitivity reactions

Quinolones have been shown to cause photosensitivity reactions in patients. However, studies have shown that moxifloxacin has a lower risk to induce photosensitivity. Nevertheless patients should be advised to avoid exposure to either UV irradiation or extensive and/or strong sunlight during treatment with moxifloxacin.

Patients with glucose-6-phosphate dehydrogenase deficiency

Patients with a family history of or actual glucose-6-phosphate dehydrogenase deficiency are prone to haemolytic reactions when treated with quinolones. Therefore, moxifloxacin should be used with caution in these patients.

Peri-arterial tissue inflammation

Moxifloxacin solution for infusion is for intravenous administration only. Intra-arterial administration should be avoided since preclinical studies demonstrated peri-arterial tissue inflammation following infusion by this route.

Patients with special cSSSI

Clinical efficacy of moxifloxacin in the treatment of severe burn infections, fasciitis and diabetic foot infections with osteomyelitis has not been established.

Patients on sodium diet

This medicinal product contains 787 mg (approximately 34 mmol) sodium per dose. To be taken into consideration by patients on a controlled sodium diet.

Interference with biological tests

Moxifloxacin therapy may interfere with the *Mycobacterium* spp. culture test by suppression of mycobacterial growth causing false negative results in samples taken from patients currently receiving moxifloxacin.

Patients with MRSA infections

Moxifloxacin is not recommended for the treatment of MRSA infections. In case of a suspected or confirmed infection due to MRSA, treatment with an appropriate antibacterial agent should be started.

Paediatric population

Due to adverse effects on the cartilage in juvenile animals the use of moxifloxacin in children and adolescents < 18 years is contraindicated.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department. Interaction with other medicinal products and other forms of interaction

Interactions with medicinal products

Interactions with medicinal products

An additive effect on QT interval prolongation of moxifloxacin and other medicinal products that may prolong the QTc interval cannot be excluded. This might lead to an increased risk of ventricular arrhythmias, including torsade de pointes. Therefore, co-administration of moxifloxacin with any of the following medicinal products is contraindicated:

- anti-arrhythmics class IA (e.g. quinidine, hydroquinidine, disopyramide)
- anti-arrhythmics class III (e.g. amiodarone, sotalol, dofetilide, ibutilide)
- antipsychotics (e.g. phenothiazines, pimozide, sertindole, haloperidol, sultopride)
- tricyclic antidepressive agents
- certain antimicrobial agents (saquinavir, sparfloxacin, erythromycin IV, pentamidine, antimalarials particularly halofantrine)
- certain antihistaminics (terfenadine, astemizole, mizolastine)
- others (cisapride, vincamine IV, bepridil, diphemanil).

Moxifloxacin should be used with caution in patients who are taking medication that can reduce potassium levels (e.g. loop and thiazide-type diuretics, laxatives and enemas [high doses], corticosteroids, amphotericin B) or medication that is associated with clinically significant bradycardia.

Warfarin

No interaction during concomitant treatment with warfarin on pharmacokinetics, prothrombin time and other coagulation parameters has been observed. Changes in INR (International Normalized Ratio): Cases of increased anticoagulant activity have been reported in patients receiving anticoagulants concurrently with antibiotics, including moxifloxacin. The infectious disease (and its accompanying inflammatory process), age and general status of the patient are risk factors. Although an interaction between

moxifloxacin and warfarin was not demonstrated in clinical trials, INR monitoring should be performed and, if necessary, the oral anticoagulant dosage should be adjusted as appropriate.

Digoxin

The pharmacokinetics of digoxin are not significantly influenced by moxifloxacin and vice versa. After repeated dosing in healthy individual moxifloxacin increased C_{max} of digoxin by approximately 30 % at steady state without affecting AUC or trough levels.

Pregnancy and lactation

Pregnancy

The safe use of moxifloxacin in human pregnancy has not been established. Reversible joint injuries are described in children receiving some fluoroquinolones; however, this effect has not been reported as occurring on exposed foetuses. The potential risk for humans is unknown. Consequently, the use of moxifloxacin during pregnancy is contraindicated.

Lactation

As with other fluoroquinolones, moxifloxacin has been shown to cause lesions in the cartilage of the weight bearing joints of immature animals. There is no data available in lactating or nursing women. Therefore, the use of moxifloxacin in nursing mothers is contraindicated.

Effects on ability to drive and use machines

Fluoroquinolones including moxifloxacin may result in an impairment of the patient's ability to drive or operate machinery due to CNS reactions and vision disorders (see section “*Undesirable Effects*”).

Undesirable effects

System Organ Class (MedDRA)	Common	Uncommon	Rare	Very Rare
Infections and infestations	Superinfections due to resistant bacteria or fungi e.g. oral and vaginal candidiasis			
Blood and lymphatic system disorders		Anaemia Leucopenia(s) Neutropenia Thrombocytopenia Thrombocythemia Blood eosinophilia Prothrombin time prolonged/ INR increased		Prothrombin level increased/ INR decreased Agranulocytosis
Immune system disorders		Allergic reaction	Anaphylaxis incl. very rarely life-threatening shock Allergic oedema/ angiooedema (incl. laryngeal oedema, potentially life-threatening)	

Metabolism and nutrition disorders		Hyperlipidemia	Hyperglycemia Hyperuricemia	Hypoglycemia
Psychiatric disorders*		Anxiety reactions Psychomotor hyperactivity/ agitation	Emotional lability Depression (in very rare cases potentially culminating in self-injurious behaviour, such as suicidal ideations/ thoughts, or suicide attempts) Hallucination	Depersonalization Psychotic reactions (potentially culminating in self-injurious behaviour, such as suicidal ideations/ thoughts, or suicide attempts)
Nervous system disorders*	Headache Dizziness	Par- and Dysaesthesia Taste disorders (incl. ageusia in very rare cases) Confusion and disorientation Sleep disorders (predominantly insomnia) Tremor Vertigo Somnolence	Hypoaesthesia Smell disorders (incl. anosmia) Abnormal dreams Disturbed coordination (incl. gait disturbances, esp. due to dizziness or vertigo) Seizures incl. grand mal convulsions Disturbed attention Speech disorders Amnesia Peripheral neuropathy and polyneuropathy	Hyperaesthesia
Eye disorders*		Visual disturbances incl. diplopia and blurred vision (especially in the course of CNS reactions)	Photophobia	Transient loss of vision (especially in the course of CNS reactions) Uveitis and bilateral acute iris transillumination
Ear and labyrinth disorders*			Tinnitus Hearing impairment incl. deafness (usually reversible)	
Cardiac disorders	QT prolongation in patients with hypokalaemia	QT prolongation Palpitations Tachycardia Atrial fibrillation Angina pectoris	Ventricular tachyarrhythmias Syncope (i.e., acute and short lasting loss of consciousness)	Unspecified arrhythmias Torsade de Pointes Cardiac arrest
Vascular disorders		Vasodilatation	Hypertension Hypotension	Vasculitis

Respiratory, thoracic and mediastinal disorders		Dyspnea (including asthmatic conditions)		
Gastrointestinal disorders	Nausea Vomiting Gastrointestinal and abdominal pains Diarrhoea	Decreased appetite and food intake Constipation Dyspepsia Flatulence Gastritis Increased amylase	Dysphagia Stomatitis Antibiotic-associated colitis (incl. pseudo-membranous colitis, in very rare cases associated with life-threatening complications)	
Hepatobiliary disorders	Increase in transaminases	Hepatic impairment (incl. LDH increase) Increased bilirubin Increased gamma-glutamyl-transferase Increase in blood alkaline phosphatase	Jaundice Hepatitis (predominantly cholestatic)	Fulminant hepatitis potentially leading to life-threatening liver failure (incl. fatal cases)
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria Dry skin		Bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis (potentially life-threatening)
Musculoskeletal and connective tissue disorders*		Arthralgia Myalgia	Tendonitis Muscle cramp Muscle twitching Muscle weakness	Tendon rupture Arthritis Muscle rigidity Exacerbation of symptoms of myasthenia gravis
Renal and urinary disorders		Dehydration	Renal impairment (incl. increase in BUN and creatinine) Renal failure	
General disorders and administration site conditions*	Injection and infusion site reactions	Feeling unwell (predominantly asthenia or fatigue) Painful conditions (incl. pain in back, chest, pelvic and extremities) Sweating Infusion site (thrombo-) phlebitis	Oedema	

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendonitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with

paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of quinolones and fluoroquinolones in some cases irrespective of pre-existing risk factors (*see section Warnings and Precautions*).

In isolated instances, some serious adverse drug reactions may be long-lasting (> 30 days) and disabling: such as tendinitis, tendon rupture, musculoskeletal disorders, and other reactions affecting the nervous system including psychiatric disorders and disturbances of senses.

The following undesirable effects have a higher frequency category in the subgroup of IV treated patients with or without subsequent oral therapy:

Common:	Increased gamma-glutamyl-transferase
Uncommon:	Ventricular tachyarrhythmias, Hypotension, Edema, Antibiotic associated colitis (in very rare cases associated with life threatening complications), Seizures of various clinical manifestations (incl. grand mal convulsions), Hallucinations, Renal impairment and Renal failure (due to dehydration esp. in elderly with pre-existing renal disorders)

Post Marketing Experience

Exacerbation of symptoms of myasthenia gravis.

Overdose

No specific countermeasures after accidental overdose are recommended. In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation. Concomitant administration of charcoal with a dose of 400 mg oral or intravenous moxifloxacin will reduce systemic availability of the drug by more than 80% or 20% respectively. The use of charcoal early during absorption may be useful to prevent excessive increase in the systemic exposure to moxifloxacin in cases of oral overdose

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Quinolone antibacterials, fluoroquinolones, ATC code: J01MA14

Mode of action

Moxifloxacin inhibits bacterial Type II topoisomerases (DNA gyrase and topoisomerase IV) that are required for bacterial DNA replication, transcription and repair.

PK/PD

Fluoroquinolones exhibit a concentration dependent killing of bacteria. Pharmacodynamic studies of fluoroquinolones indicate that the primary determinant of efficacy is the AUC₂₄/MIC ratio.

Mechanism of resistance

Resistance to fluoroquinolones can arise through mutations in DNA gyrase and topoisomerase IV. Other mechanisms may include over-expression of efflux pumps, impermeability, and protein-mediated protection of DNA gyrase. Cross resistance should be expected between moxifloxacin and other fluoroquinolones.

The activity of moxifloxacin is not affected by mechanisms of resistance that are specific to antibacterial agents of other classes.

Breakpoints

EUCAST clinical MIC and disk diffusion breakpoints for moxifloxacin (01.01.2012):

Organism	Susceptible	Resistant
<i>Staphylococcus</i> spp.	≤ 0.5 mg/l ≥ 24 mm	> 1 mg/l < 21 mm
<i>S. pneumoniae</i>	≤ 0.5 mg/l ≥ 22 mm	> 0.5 mg/l < 22 mm
<i>Streptococcus</i> Groups A, B, C, G	≤ 0.5 mg/l ≥ 18 mm	> 1 mg/l < 15 mm
<i>H. influenzae</i>	≤ 0.5 mg/l ≥ 25 mm	> 0.5 mg/l < 25 mm
<i>M. catarrhalis</i>	≤ 0.5 mg/l ≥ 23 mm	> 0.5 mg/l < 23 mm
<i>Enterobacteriaceae</i>	≤ 0.5 mg/l ≥ 20 mm	> 1 mg/l < 17 mm
Non-species related breakpoints*	≤ 0.5 mg/l	> 1 mg/l

* Non-species related breakpoints have been determined mainly on the basis of pharmacokinetic/pharmacodynamic data and are independent of MIC distributions of specific species. They are for use only for species that have not been given a species-specific breakpoint and are not for use with species where interpretative criteria remain to be determined.

Microbiological Susceptibility

The prevalence of acquired resistance may vary geographically and with time for selected species and local information of resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought where the local prevalence of resistance is such that utility of the agent in at least some types of infections is questionable.

Commonly susceptible species
<u>Aerobic Gram-positive micro-organisms</u> <i>Staphylococcus aureus</i> * ⁺ <i>Streptococcus agalactiae</i> (Group B) <i>Streptococcus milleri</i> group* (<i>S. anginosus</i> , <i>S. constellatus</i> and <i>S. intermedius</i>) <i>Streptococcus pneumoniae</i> * <i>Streptococcus pyogenes</i> * (Group A) <i>Streptococcus viridans</i> group (<i>S. viridans</i> , <i>S. mutans</i> , <i>S. mitis</i> , <i>S. sanguinis</i> , <i>S. salivarius</i> , <i>S. thermophilus</i>)
<u>Aerobic Gram-negative micro-organisms</u> <i>Acinetobacter baumannii</i> <i>Haemophilus influenzae</i> * <i>Legionella pneumophila</i> <i>Moraxella</i> (<i>Branhamella</i>) <i>catarrhalis</i> *
<u>Anaerobic micro-organisms</u> <i>Prevotella</i> spp.

<u>“Other” micro-organisms</u> <i>Chlamydomphila (Chlamydia) pneumoniae</i> * <i>Coxiella burnetii</i> <i>Mycoplasma pneumoniae</i> *
Species for which acquired resistance may be a problem
<u>Aerobic Gram-positive micro-organisms</u> <i>Enterococcus faecalis</i> * <i>Enterococcus faecium</i> *
<u>Aerobic Gram-negative micro-organisms</u> <i>Enterobacter cloacae</i> * <i>Escherichia coli</i> *# <i>Klebsiella oxytoca</i> <i>Klebsiella pneumoniae</i> *# <i>Proteus mirabilis</i> *
<u>Anaerobic micro-organisms</u> <i>Bacteroides fragilis</i> *
Inherently resistant organisms
<u>Aerobic Gram-negative micro-organisms</u> <i>Pseudomonas aeruginosa</i>
*Activity has been satisfactorily demonstrated in clinical studies. +Methicillin resistant <i>S. aureus</i> have a high probability of resistance to fluoroquinolones. Moxifloxacin resistance rate of > 50% have been reported for methicillin resistant <i>S. aureus</i>. #ESBL-producing strains are commonly also resistant to fluoroquinolones.

PHARMACOKINETICS

Absorption:

After a single 400 mg intravenous 1 h infusion peak concentrations of approximately 4.1 mg/L were reached in the plasma at the end of infusion which corresponds to a mean increase of approx. 26% relative to the oral application. Exposure to drug in terms of AUC at a value of approximately 39 mg/L is only slightly higher compared to the exposure after oral administration (35 mg/L) in accordance with the absolute bioavailability of approximately 91%. Following multiple intravenous dosing (1 h infusion), peak and trough plasma concentrations at steady state (400 mg once daily) were between 4.1 to 5.9 and 0.43 to 0.84 mg/L respectively. At steady-state the exposure to drug within the dosing interval is approximately 30% higher than after the first dose. In patients mean steady state concentrations of 4.4mg/L were observed at the end of a 1 h infusion.

Distribution

Moxifloxacin is distributed very rapidly to extravascular spaces. Exposure to drug in terms of AUC (AUC_{norm} = 6 kg*h/L) is high with a volume of distribution at steady state (V_{ss}) of approx. 2 L/kg. In ex vivo experiments over a range of 0.02 to 2 mg/L a protein binding of approximately 45 % independent from the concentration of the drug was determined. Moxifloxacin is mainly bound to serum albumin. Due to this low value high free peak concentrations > 10 x MIC are observed. Moxifloxacin reaches high concentrations in tissues like lung (epithelial fluid, alveolar macrophages, biotic tissue), the sinuses (maxillary and ethmoid sinus, nasal polypi) and inflamed lesions (cantharide blister fluid) where total concentrations exceeding those of the plasma concentrations are reached. In addition, high drug

concentrations were detected in abdominal tissues and fluids and female genital tract. The peak concentrations and site vs. plasma concentration ratios for various target tissues yielded comparable results for drug administration after a single dose of 400 mg moxifloxacin.

Metabolism

Moxifloxacin undergoes Phase II biotransformation and is excreted via renal and biliary/faecal pathways as unchanged drug as well as in form of a sulfa-compound (M1) and a glucuronide (M2). M1 and M2 are the only metabolites relevant in humans, both are microbiologically inactive. Independent from the route of administration the metabolites M1 and M2 are found in the plasma at concentrations lower than the parent drug. Preclinical investigations adequately covered both metabolites thus excluding potential implications with respect to safety and tolerability.

Elimination

Moxifloxacin is eliminated from plasma with a mean terminal half-life of approximately 12 hours. The mean apparent total body clearance following a 400 mg dose ranges from 179 to 246 mL/min. Renal clearance amounted to about 24 - 53 mL/min suggesting partial tubular reabsorption of the drug from the kidneys. Concomitant administration of ranitidine and probenecid did not alter renal clearance of the drug. Mass balance of the mother compound and Phase II metabolites of moxifloxacin yielded an almost complete recovery of approx. 96-98% independent from the route of administration with no indication of oxidative metabolism.

Elderly

Pharmacokinetics of moxifloxacin is not affected by age.

PHARMACEUTICAL PARTICULARS

Incompatibilities

The following solutions are incompatible with MOXICUM solution for infusion:

Sodium chloride 10% and 20% solutions

Sodium bicarbonate 4.2% and 8.4% solutions

This medicinal product must not be mixed with other medicinal products.

If MOXICUM solution for infusion is to be given with another drug, each drug should be given separately. Only clear solutions are to be used.

Special precautions for storage

Store below 30°C and protect from light. Avoid extreme heat and freezing. Keep out of reach of children.

This product is for single use only. Any unused solution should be discarded.

Nature and contents of container

250 ml Type II borosilicate colourless glass vial sealed with a bromobutyl rubber stopper, aluminium seal ring and flip off cap. Pack size is box containing 1 vial.

Special precautions for disposal and other handling

This product is for single use only. Any unused solution should be discarded.

The following co-infusions were found to be compatible with moxifloxacin 400 mg solution for infusion:

Water for injections, Sodium chloride 0.9%, Sodium chloride 1 molar, Glucose 5%/10%/40%, Xylitol 20%, Ringer's solution, Compound Sodium Lactate Solution (Hartmann's Solution, Ringer-Lactate Solution).

Moxifloxacin solution for infusion should not be co-infused with other drugs.

Do not use if there are any visible particulate matter or if the solution is cloudy.

Manufacturer

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